LowSYS — Status Report

Distillation column simulator (Flutter: Windows + Android) · 12 Oct 2026, 6:10 PM MT · repo: github.com/HearthOfEmbers/LowSYS (private)

3 of 10	**292**	**60 fps**	**≈1e-12**
milestones built (M1–M3)	validation tests passing (Tier 1)	Windows, 100× sim speed	mass-balance closure

Milestones

#	Scope	Status
M1	Dynamic binary column (textbook thermo), animated column, live McCabe-Thiele, Blueprint theme	Done
M2	Tray hydraulics (ΔP, flooding, weeping/dumping, entrainment, downcomer), faults F01–F07, tap-for-equations overlay, SI/US units, scenarios G11–G12	Done
M3	PID loops (MAN/AUTO), instruments with noise, instrument/valve faults, alarms, balance cross-checks, trends, save/load, scenarios G1–G6	Done*
M4	Rigorous thermo + energy balances, pressure dynamics, condenser/reboiler faults, Control Room theme	Spec ready, tests written
M5	Non-ideal thermo (NRTL), azeotropes (ethanol–water, acetone–chloroform), T-x-y, Paper theme	Spec partial
M6	Thermosyphon reboiler, hidden-fault drills (incl. your instrument-failure drills H9/H10)	Specced
M7–M10	Multicomponent hydrocarbons, packed columns, crude fractionation, polish/iOS/web	Specced

* Remaining M3 items (see DEVIATIONS.md "M3 handoff"): verify frame-time fix, Android emulator pass, minor UI polish.

What's been verified

- Engine matches independent reference solvers: default column x_D 0.981183 / x_B 0.018817 to 1e-7; 13 steady-state cases; results identical at time steps 0.25–30 s.
- Hydraulics match Fair / Bennett / Kister rules of thumb; default column now runs at 63% of flood (was 83%) so "more reflux = better" holds on the first slider pull.
- Rigorous-thermo data cross-checked against NIST WebBook; Tier-2 reference steady state and energy closure confirmed ahead of coding.
- Instrument-failure drills checked statistically: faults stand out 2.7–7.5× above noise (phone-winnable).

Decisions made on your behalf

- Tray spacing 0.90 m for flood headroom · dumping trays drain to ~0.15 mm (physically honest) · G8 pressure sweep starts at 0.7 bar (real condenser limit).
- Noise on in sandbox, off in guided scenarios · H9 includes a "nothing is broken" decoy · Smart App Control turned off (by you) to allow builds.

Next steps (automatic)

- **9:45 PM MT tonight:** scheduled task on SHIVA starts a fresh coordinator from `docs/STATUS.md` (Claude app must be open).
- Apply 7 queued spec fixes → build M4 (fresh Sonnet) → validate M4 (Opus xhigh, key numbers only).
- Usage-saving rules now in force: fresh agents per task, just-in-time specs, per-milestone doc excerpts, fewer screenshots.

Open for you (non-blocking): a real crude assay for the crude-unit preset, whenever you find one.